

Aile Ma

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EDUCATION

Yale University, New Haven, CT Sep 2024 - Jun 2026
MS in Biostatistics (Data science pathway)

University of Washington, Seattle, WA (Grade: 3.8 / 4.0 | UW Annual & Quarterly Dean's List) Sep 2020 - Jun 2024
BS in Statistics (Data Science pathway) & Applied Mathematics

Relevant Coursework: Applied Regression & ANOVA, High-Performance Computing, Computational Data Analysis, Database Systems, Vector Calculus, Data Visualization, Advanced Regression, SAS & R Programming, Optimization, Probability Theory, Machine Learning, Deep Learning, Longitudinal & Survival Analysis.

INTERNSHIP EXPERIENCE

Novartis, East Hanover, NJ Jun 2025 - Aug 2025
Data Science Intern – Strategy and Analytics (Project Coordinator)

- Assisted in developing SQL queries in Snowflake to prepare and optimize data for Power BI reporting for the Cosentyx team within Novartis Patient Support (NPS).
- Supported the creation of an automated Power BI dashboard to track Paid & Bridge first-fill and triage rates by payer, access line, and pharmacy, replacing manual slide preparation.
- Reduced reporting prep time by ~3 workdays/month (~300 hrs/year) and enabled real-time KPI monitoring, enhancing decision-making for payer/access strategies and adherence program effectiveness.

Allied Stone Inc, Dallas, TX Jul 2023 - Sep 2023 & Jul 2022 - Sep 2022
Business Intelligence Analyst Intern

- Designed and deployed dynamic dashboards in Domo for real-time tracking of cabinet performance metrics.
- Optimized data workflows by implementing ETL processes using Python (pandas) and SQL; cleaned and analyzed data for enhanced dashboard visuals.
- Streamlined ERP automation and cross-system analytics, reducing manual processing time by 30% and improving operational reporting efficiency.

RESEARCH EXPERIENCE

Lake Lab (Yale) Nov 2024 - Now
Research Assistant

- Built a genomic data pipeline to identify and annotate mitochondrial and nuclear ORFs, applying codon mapping, frame analysis, and circular genome handling.
- Created algorithms for start/stop codon detection and dual-code sequence translation, validated against NCBI datasets to resolve annotation discrepancies.
- Analyzed mitochondrial vs. nuclear ORF properties, producing comparative visualizations and conservation metrics to prioritize candidates for further study.

Class-Conditional Diffusion Model for Fashion-MNIST (Yale) Sep 2024 - Dec 2024
Researcher

- Trained a Conditional Diffusion Model using PyTorch and Diffusers to generate class-conditioned Fashion-MNIST images, optimizing the noise predictor with Adam over 20 epochs for stable training and improved generation quality.
- Implemented Noise Scheduling with DDPM Scheduler to improve training stability and enhance image generation quality.
- Evaluated Model Performance by analyzing loss convergence, visualizing synthetic images, and comparing generated samples across different class labels, showcasing the model's ability to produce diverse and realistic outputs.

New York Taxi Trip Duration Prediction (MIT) Mar 2022 - Jun 2022
Team Lead

- Spearheaded the development of a predictive pipeline for New York taxi trip durations, leveraging ensemble learning (XGBoost) and regression models in scikit-learn to achieve optimized performance metrics (MAE, MSE, RMSE).
- Engineered spatiotemporal features from geospatial clustering, traffic patterns, and distance metrics to boost model accuracy.
- Created visualizations in Seaborn/Matplotlib to uncover non-linear trends in ride duration and congestion patterns.

SKILLS AND INTERESTS

- *ML Algorithms:* Logistic & linear Regression, Decision Trees, Random Forest, XGboost, PCA, K-means, HC
- *Deep Learning:* ANN, CNN, LSTM/RNN, Transfer Learning, NLP
- *Programming Languages:* Python, R, SQL, SAS, MATLAB, Tableau, Power BI
- *Statistical Methods:* Distributions, Bayes, Inferential Statistics, ANOVA, Survival Analysis
- *Visualisation:* ggplot2, Seaborn, Matplotlib, Shiny